

Being Socially Aware in Technology Design

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CCS CONCEPTS

• **Human-Centered Computing, Human-Computer Interaction (HCI), Interaction Design, Collaborative and Social Computing;**

KEYWORDS

computational technology design, societal issues in computing, intersubjective aspects of interaction

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In the last decade, several initiatives of the Human-Computer Interaction (HCI) community have brought about discussion on different matters surrounding social issues in design. The addressed topics range from considering political issues of power, privilege, and access in their complexity, to the “doing good” in HCI. Moreover, the technological evolution along the history of computing, has also added complexity with social issues arising from the ubiquity of technology and the not always noticeable interaction with it; examples of situations such as inadvertent capture of individual data in smart environments, and the combination of data with sophisticated machine learning algorithms, are already well-known issues for considerations on responsibilities. Thus, designing computer-based artefacts necessarily involves making a number of implicit and explicit epistemological and ontological assumptions, which affect the system in terms of its design, and impact of its use, reflecting a world view.

In my view, being socially aware in design goes beyond having real people informing design, paying attention to the social implications of computer technology within underserved societies, or being an activist of social causes related to technology, although all these are relevant approaches. It has also to do with the meaning of being human and understanding the nature of human action in the world as part of the process of creating computational systems. Thus, we acknowledge that there is no neutrality in our relationship with technology: we suffer the impact of technology as part of society and we are responsible for the form it takes and the effects it causes, as designers, developers and interested parties in technology-based systems.

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I think of technology as an emancipatory vehicle, with opportunities of (social) growth for people participating in its creation. As so, design is understood as a social process of joint construction of meaning that manifests itself in a computer-based product that mediates our interaction in the world. This understanding leads to a number of concepts related to effective participation of interested parties in the process of design, which are rooted in how we understand a computational system, how we situate it in a social context, to create products that make sense in it [1–3].

Recognizing the situational nature of design, this talk invites a reflection on the social and intersubjective aspects of the experience that should be made explicit in technology-based design. A socially aware perspective on technology design (Socially Aware Design or SAwD for short) has evolved over different academic works and research projects conducted in the Brazilian context along the last 20 years. It has been instantiated in various social scenarios such as: in academia [9], industry [4, 5], civil society [7, 8, 10], education [6].

In sum, I like to think of design, not as a situation in which some people are inventing technology for (other) people to use, rather as people inventing new ways of living in scenarios that are created with technology. Thus, designing technology means designing (social) scenarios in which technology is embedded and life is lived. In this sense, being socially aware of the different design environments, technologies and ways they afford people’s interaction, is essential. For this, a coherent view of responsibilities in the creation of technology and a collection of artifacts, techniques, methods and attitude are foundations for acting out a better human-technology-society relation.

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